

Refrigerator & freezer standards are changing.

A two-phase compliance transition under 10 CFR 430.32. Current 2014 limits stay in force until the dates below, by product class.

JANUARY 31, 2029

Built-in products
Compact products
Chest freezers
Selected special classes
including new class 9A-BI

JANUARY 31, 2030

Freestanding standard
refrigerator-freezers
Top / side / bottom mount
All-refrigerators
Upright auto-defrost freezers

What else is changing

● Tighter kWh/yr limits

Maximum annual energy use is calculated from adjusted volume (AV).

● Icemaker factor I

No separate icemaker classes. I = 1 adds 28 kWh/yr; I = 0 if none.

● Door coefficients K

Transparent door +10%. Door-in-door +6%. Extra external doors +2% each.

● National impact

DOE estimates ~11% energy savings vs. no-new-standards, or 5.6 quads over 30 years.

Who must comply when

Dates apply to products manufactured in, or imported into, the United States.

Date	Product group	Product classes
Jan 31, 2029	Built-in refrigerator-freezers and freezers	3-BI, 3A-BI, 4-BI, 5-BI, 5A-BI, 7-BI, 9-BI, 9A-BI
Jan 31, 2029	Bottom-mount with through-the-door ice (freestanding)	5A
Jan 31, 2029	Upright manual-defrost freezers	8
Jan 31, 2029	Chest freezers (except compact)	10, 10A
Jan 31, 2029	Compact refrigerators, refrigerator-freezers, and freezers	11, 11A, 12, 13, 13A, 14, 15, 16, 17, 18
Jan 31, 2030	Freestanding standard refrigerator-freezers	1, 1A, 2, 3, 3A, 4, 5, 6, 7
Jan 31, 2030	Freestanding upright auto-defrost freezers	9

Until then

Products still follow the September 15, 2014 standards in Table 1 to 10 CFR 430.32(a)(1) until their class-specific 2029 or 2030 date. Limits do not apply to refrigerators / refrigerator-freezers over 39 ft³ or freezers over 30 ft³.

TABLE 2 · 10 CFR 430.32(A)(2)

Maximum energy use from Jan 31, 2029

kWh/yr from adjusted volume AV (ft³). I = 1 with automatic icemaker. K = door coefficient.

Class	Product	Max energy use (kWh/yr)
3-BI	Built-in RF, top freezer	$8.24AV + 238.4 + 28I$
3A-BI	Built-in all-refrigerator, auto defrost	$(7.22AV + 205.7) \times K3ABI$
4-BI	Built-in RF, side freezer	$(8.79AV + 307.4) \times K4BI + 28I$
5-BI	Built-in RF, bottom freezer	$(8.65AV + 309.9) \times K5BI + 28I$
5A	RF, bottom freezer + through-the-door ice	$(7.76AV + 351.9) \times K5A$
5A-BI	Built-in RF, bottom freezer + TTD ice	$(8.21AV + 370.7) \times K5ABI$
7-BI	Built-in RF, side freezer + TTD ice	$(8.82AV + 384.1) \times K7BI$
8	Upright freezer, manual defrost	$5.57AV + 193.7$
9-BI	Built-in upright freezer, auto defrost	$(9.37AV + 247.9) \times K9BI + 28I$
9A-BI	Built-in upright freezer, auto defrost + TTD ice	$9.86AV + 288.9$
10	Chest freezer (except compact)	$7.29AV + 107.8$
10A	Chest freezer, auto defrost	$10.24AV + 148.1$
11	Compact RF / refrigerator, manual defrost	$7.68AV + 214.5$
11A	Compact all-refrigerator, manual defrost	$6.66AV + 186.2$
12	Compact RF, partial auto defrost	$(5.32AV + 302.2) \times K12$
13	Compact RF, top freezer, auto defrost	$10.62AV + 305.3 + 28I$
13A	Compact all-refrigerator, auto defrost	$(8.25AV + 233.4) \times K13A$
14	Compact RF, side freezer, auto defrost	$6.14AV + 411.2 + 28I$
15	Compact RF, bottom freezer, auto defrost	$10.62AV + 305.3 + 28I$
16	Compact upright freezer, manual defrost	$7.35AV + 191.8$
17	Compact upright freezer, auto defrost	$9.15AV + 316.7$
18	Compact chest freezer	$7.86AV + 107.8$

Maximum energy use from Jan 31, 2030

Freestanding standard classes. kWh/yr from AV (ft³). I = 1 with automatic icemaker.

Class	Product	Max energy use (kWh/yr)
1	RF / refrigerators other than all-refrigerators, manual defrost	$6.79AV + 191.3$
1A	All-refrigerators, manual defrost	$5.77AV + 164.6$
2	RF, partial automatic defrost	$(6.79AV + 191.3) \times K2$
3	RF, auto defrost, top-mounted freezer	$6.86AV + 198.6 + 28I$
3A	All-refrigerators, automatic defrost	$(6.01AV + 171.4) \times K3A$
4	RF, auto defrost, side-mounted freezer	$(7.28AV + 254.9) \times K4 + 28I$
5	RF, auto defrost, bottom-mounted freezer	$(7.61AV + 272.6) \times K5 + 28I$
6	RF, auto defrost, top freezer + TTD ice	$7.14AV + 280.0$
7	RF, auto defrost, side freezer + TTD ice	$(7.31AV + 322.5) \times K7$
9	Upright freezers, automatic defrost	$(7.33AV + 194.1) \times K9 + 28I$

How to read the equation

- AV = total adjusted volume in cubic feet (appendices A and B to subpart B).
- I = 1 if the product has an automatic icemaker; I = 0 if it does not.
- K2, K3A, K4, K5, K7, K9 are door coefficients from Table 5 (next slide).
- Round the result to the nearest kWh/yr; round .5 upward.

K factors for special doors

These multipliers raise the allowed kWh/yr for transparent doors, door-in-door, and extra doors.

How the allowance works

Transparent door K = 1.10 (+10%)	Door-in-door K = 1.06 (+6%)	Each extra external door +2% (formula with Nd)
--	---------------------------------------	--

2029 classes · Table 3

K	Transparent door	Door-in-door	Added external doors
K3ABI	1.10	1.00	1.00
K4BI / K5BI / K7BI	1.10	1.06	1 + 0.02 × (Nd − 2)
K5A / K5ABI	1.10	1.06	1 + 0.02 × (Nd − 3)
K9BI / K12	1.00	1.00	1 + 0.02 × (Nd − 1)
K13A	1.10	1.00	1.00

2030 classes · Table 5

K	Transparent door	Door-in-door	Added external doors
K2	1.00	1.00	1 + 0.02 × (Nd − 1)
K3A	1.10	1.00	1.00
K4 / K5 / K7	1.10	1.06	1 + 0.02 × (Nd − 2)
K9	1.00	1.00	1 + 0.02 × (Nd − 1)

Nd = number of external doors. Maximum Nd: 2 for K2 and K12; 3 for K9BI; 5 for all other K values.